

Sequences and Their Limits

Benjamin Abshire[†]
Kent State University

October 7, 2025

I Introduction

Definition. A sequence $\{x_i\}_{i=1}^{\infty}$ is convergent if:

$$\exists x \in \mathbb{R} \text{ s.t. } \forall \epsilon > 0, \exists N_0 : n \geq N_0 \text{ s.t. } |x - x_n| < \epsilon.$$

Remark. If such x exists, it is unique, and x is the limit of $\{x_i\}_{i=1}^{\infty}$. i.e.

$$\lim_{n \rightarrow \infty} x_i = x$$

Theorem. Consider the sequence $\{x_n\}_{n=1}^{\infty}$. Assume there exists an $\{a_n\}$ and a number x such that:

$$\lim_{n \rightarrow \infty} a_n = 0 \text{ and}$$

Then, $\lim_{n \rightarrow \infty} x_n = x$.

Proof.

$$a_n \rightarrow 0. \text{ For all } \epsilon > 0, \exists N_0(\epsilon) \text{ s.t. } \forall n > N_0(\epsilon), |a_n| < \epsilon \implies a_n < \epsilon \text{ but } |x_n - x| \leq a_n < \epsilon$$

Therefore, $\lim_{n \rightarrow \infty} x_n = x$ $\square$

Theorem (Squeeze Theorem). Suppose that x_n , y_n , and z_n are sequences of real numbers such that

$$x_n \leq z_n \leq y_n, \forall n \in \mathbb{N}$$

and $\lim x_n = \lim y_n$. Then, z_n is convergent and

$$\lim x_n = \lim z_n = \lim y_n$$

[†]Definitions and example problems from MATH 42001: Analysis I instructed by Dr. Artem Zvavitch and *Introduction to Real Analysis* by Robert G. Bartle and Donald R. Sherbert

Proof. Let $\omega = \lim x_n = \lim z_n$.

$\forall \epsilon > 0, \exists N_1 : \forall n > N_1, |y_n - \omega| < \epsilon$. By definition, $-\epsilon < y_n - \omega < \epsilon$

$$\implies y_n < \epsilon + \omega \implies x_n \leq y_n < \epsilon + \omega.$$

We now want to show that z_n is convergent and in between x_n and y_n

$\exists N_2 : \forall n > N_2, |x_n - \omega| < \epsilon$. By definition, $\omega - \epsilon < x_n < z_n$. So, $\omega - \epsilon < z_n < \omega + \epsilon$.

Note that $n > \max\{N_1, N_2\}$. $\square$

Now we will state some other theorems that will help us prove the convergence or divergence of any sequence.

Theorem. $x_n > 0$

$$\lim_{n \rightarrow \infty} x_n = x \implies \lim_{n \rightarrow \infty} \sqrt{x_n} = \sqrt{x}$$

Theorem. If $\lim_{n \rightarrow \infty} \frac{x_n}{x_{n+1}} = L < 1$ then,

$$\lim_{n \rightarrow \infty} x_n = 0$$

Theorem. $\lim_{n \rightarrow \infty} x_n = x \implies \lim_{n \rightarrow \infty} |x_n| = |x|$

Theorem. $\lim_{n \rightarrow \infty} x_n = x \implies \lim_{n \rightarrow \infty} x_n^k = x^k$, where $k \in \mathbb{N}$

II Examples

1. Prove the convergence or divergence of $\{x_n\}_{n=1}^{\infty} = \frac{1}{n}$

Proof. We know from calculus that

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

We want to show that for any $\epsilon > 0$, we can find $N_0(\epsilon)$ (a starting point) so that $n > N_0(\epsilon)$.

$$\left| \frac{1}{n} - 0 \right| < \epsilon \implies \frac{1}{n} < \epsilon \implies n > \frac{1}{\epsilon}.$$

By the Archimedean Property,

$$\exists N_0(\epsilon) > \frac{1}{\epsilon}. \text{ Then, if } n > N_0(\epsilon), n > \frac{1}{\epsilon}.$$

Therefore, the sequence converges to zero. $\square$

2. Prove the convergence or divergence of $\{x_n\}_{n=1}^{\infty} = \frac{1}{n^2 + 1}$

Proof. For all $\epsilon > 0$, we know that:

$$\frac{1}{n^2 + 1} \leq \frac{1}{n^2} \leq \frac{1}{n}$$

By the definition of convergence:

$$\left| \frac{1}{n^2 + 1} - 0 \right| = \frac{1}{n^2 + 1} \leq \frac{1}{n} < \epsilon$$

By our result in problem 1, we know that $\frac{1}{n}$ converges to zero. Since $\frac{1}{n^2 + 1} \leq \frac{1}{n}$, then $\frac{1}{n^2 + 1}$ also converges to zero. $\square$

3. Prove the convergence or divergence of $\{x_n\}_{n=1}^{\infty} = \frac{2(-1)^n}{2 - n}$

Proof. We know from calculus that

$$\lim_{n \rightarrow \infty} \frac{2(-1)^n}{2 - n} = 0$$

By the definition of convergence:

$$\left| \frac{2(-1)^n}{2 - n} - 0 \right| < \epsilon \implies \frac{2}{|2 - n|} < \epsilon \implies n > \frac{2}{\epsilon} + 2$$

By the Archimedean Property,

$$\exists N_0(\epsilon) > \frac{2}{\epsilon} + 2. \text{ Then, if } n > N_0(\epsilon) \implies n > \frac{2}{\epsilon} + 2$$

Therefore, the sequence converges to zero. $\square$

4. Prove the convergence or divergence of $\frac{n+1}{n+2}$.

Proof. We know from calculus that

$$\lim_{n \rightarrow \infty} \frac{n+1}{n+2} = 1$$

By the definition of convergence:

$$\begin{aligned} \left| \frac{n+1}{n+2} - 1 \right| < \epsilon &\implies \left| \frac{n+1}{n+2} - \left(\frac{n+2}{n+2} \right) \right| < \epsilon \implies \left| \frac{n+1 - n - 2}{n+2} \right| < \epsilon \\ &\implies \left| \frac{-1}{n+2} \right| < \epsilon \implies \frac{1}{n+2} < \epsilon \end{aligned}$$

For all $\epsilon > 0$ we know that:

$$\frac{1}{n+2} < \frac{1}{n}$$

So,

$$\frac{1}{n+2} < \frac{1}{n} < \epsilon$$

Therefore, the sequences converges to 1. $\square$

5. Prove the convergence or divergence of $3 + (-1)^n$.

Proof. When n is odd, $a_n = 2$ and when n is even, $a_n = 4$. Assume a_n converges to x . Then,

$$\exists x \in \mathbb{R} \text{ s.t. } \forall \epsilon > 0, \exists N_0 : n \geq N_0 \text{ s.t. } |x - x_n| < \epsilon.$$

Take $\epsilon = 1$. Then,

$$|4 - x| < 1 \text{ and } |2 - x| < 1$$

which is a contradiction. x cannot be between 3 and 5 and between 1 and 3. Therefore, a_n does not converge, so it diverges. $\square$